

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

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I Betty A. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

19/20
Ant

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.
4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a /27 **subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q1. Subnetting for 192.168.20.0/24

Lab A:

Subnet : /26

Network Address : 192.168.20.0

Valid Host Range : 192.168.20.1 - 192.168.20.62

Broadcast Address : 192.168.20.63

Usable Hosts : 62

Lab B:

Subnet : /27

Network Address : 192.168.20.64

Valid Host Range : 192.168.20.65 - 192.168.20.94

Broadcast Address : 192.168.20.95

Usable Hosts : 30

Lab C:

Subnet : /28

Network Address : 192.168.20.96

Valid Host Range : 192.168.20.97 - 192.168.20.110

Broadcast Address : 192.168.20.111

Usable Hosts : 14

Interpretation:

The 124 network is divided into three fixed-size subnets. Each lab receives its own network, broadcast address, and host range.

Q2. Analyze the two systems

Subnet mask : 255.255.255.192 (126)

System 1:

IP Address : 192.168.1.130

Network Address : 192.168.1.128

Valid Hosts : 192.168.1.129 - 192.168.1.190

Broadcast Address : 192.168.1.191

IP belongs to this subnet : Yes

Usable Hosts : 62

System 2:

IP Address : 192.168.1.190

Network Address : 192.168.1.128

Valid Hosts : 192.168.1.129 - 192.168.1.190

Broadcast Address : 192.168.1.191

IP belongs to this subnet : Yes

Usable Hosts : 62

Interpretation :

Both systems are in the same subnet (192.168.1.128/26) and can communicate directly without a router.

Q3. Subnetting for 172.16.0.0/16

HR :

Subnet : 125

Network Address : 172.16.0.0

Valid Host Range : 172.16.0.1 - 172.16.0.126

Broadcast Address : 172.16.0.127

Usable Hosts : 126

Sales :

Subnet : 126

Network Address : 172.16.0.128

Valid Host Range : 172.16.0.129 - 172.16.0.190

Broadcast Address : 172.16.0.191

Usable Hosts : 62

IT :

Subnet : 127

Network Address : 172.16.0.192

Valid Host Range : 172.16.0.193 - 172.16.0.222

Broadcast Address : 172.16.0.223

Usable Host : 30

Admin :

Subnet : 128

Network Address : 172.16.0.224

Valid Host Range : 172.16.0.225 - 172.16.0.238

Broadcast Address : 172.16.0.239

Usable Hosts : 14

Remaining unused IP Range:

172.16.0.240 - 172.16.255.255

Interpretation:

Each department receives a separate subnet, while the remaining addresses are available for future expansion.

Q4. Subnetting for 172.16.10.0 /24 using /27

Number of subnets = 8

Usable hosts per subnet = 30

Subnets:

172.16.10.0 /27

172.16.10.32 /27

172.16.10.64 /27

172.16.10.96 /27

172.16.10.128 /27

172.16.10.160 /27

172.16.10.192 /27

172.16.10.224 /27

For 172.16.10.78

Network Address : 172.16.10.64

Valid Hosts : 172.16.10.65 - 172.16.10.94

Broadcast Address : 172.16.10.95

For 172.16.10.95

- It is the broadcast address, not a host address
- Therefore, it does not fall within the valid host range.

Interpretation:

IP 172.16.10.78 is a valid host in subnet 172.16.10.64/27, while 172.16.10.95 is reserved as the broadcast address.

Q5. IPv6 Address

Given :

2001:0db8:0000:0000:0000:ff00:0042:8329

compressed form :

2001:db8::ff00:42:8329

Expanded form :

2001:0db8:0000:0000:0000:ff00:0042:8329

Network prefix (164) :

2001:0db8:0000:0000::164

Total Host Addresses :

$$\text{Host bits} = 128 - 64 = 64$$

Total possible host addresses :

$$2^{64} = 18,446,744,073,709,551,616$$

Interpretation :

The IPv6 address is compressed by removing leading zeros and consecutive zero blocks. With a 164 prefix, the first 64 bits identify the network, while the remaining 64 bits are available for host addresses.